TEMAS

Release 0.0.1

Rubu GmbH

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API REFERENCE

1 src 1

2 temas module 3

3 Indices and tables 11

Python Module Index 13

CHAPTER

ONE

SRC

CHAPTER

TWO

TEMAS MODULE

class rubu.temas.**Connect**(*hostname='temas'*, *ip_address=''*)

Bases: object

This class alows the Temas connection.

get_ip_from_hostname(*hostname*, *max_attempts=4*)

Attempts to determine the IP address of the hostname.

class rubu.temas.**Socket**(*port*, *ip_address=''*)

Bases: object

This class send commands to Temas

send_cmd(*command*, *text=False*, *path=''*)

Send commands

Parameters
command (*List, optional*) – A command is given

Returns
Data of socket `temas.send_cmd`

Raises
Exception

class rubu.temas.**Camera**(*port=8081*, *ip_address=''*)

Bases: object

This class allows the control of camera using urllib.

update()

Continuously fetches frames from the MJPEG stream.

start_thread()

Starts the thread to fetch frames.

stop_thread()

Stops the thread.

get_frame()

Returns the most recent frame

Returns
Frame

Return type
numpy.ndarray or None

set_exposure_time(*microseconds*)

Set exposure time of visual camera 50-33000µs

set_brightness(*brightness_percent*)

Set brightness of visual camera 0-100

set_saturation(*saturation_percent*)

Set saturation of visual camera 0-100

set_contrast(*contrast_percent*)

Set contrast of visual camera 0-100

set_lens_position(*lens_position_percent*)

Set lens position of visual camera 0-100

set_gain(*gain_value*)

Set gain of visual camera 1.23-16.0

class rubu.temas.**Control**(*port=8082*, *ip_address=''*)

Bases: object

This class alows the control of motors and laser.

get_pcl(*path*)

Receive last point cloud

Returns
.ply file

Return type
String

distance()

Laser distance measurment

Returns
Value of the distance [cm]

Return type
String

mean_distance()

Mean laser distance measurement

Returns
Value of the mean_distance [cm]

Return type
String

get_focus_position()

Get focus position of visual camera 0-100

Returns
Get focus position of visual camera

Return type
String

`set_focus_position`(*position*)

Set focus position of visual camera 0-100

Returns
Set focus position of visual camera

Return type
String

`get_laser_los`()

Get laser pointer position on visual camera (x,y)

Returns
Get laser los [px,px]

Return type
String

`set_laser_los`(*x*, *y*)

Set laser pointer position on visual camera (x,y)

`move_pos`(*phi*, *theta*)

Move to coordinate

`get_pos`()

Get motor positions

Returns
pan, tilt positions [°]

Return type
String

`move_home`()

Move to home position

Returns
Motor at start position?

Return type
Boolean

`move`(*direction*)

Sends a movement command to the motor.

`move_threaded`(*direction*)

Starts the movement in a separate thread.

`move_right`()

Move motor right in 1° steps

`move_left`()

Move motor left in 1° steps

`move_up`()

Move motor up in 1° steps

`move_down`()

Move motor down in 1° steps

move_right_fine()

Move motor right in 0.5° steps

move_left_fine()

Move motor left in 0.5° steps

move_up_fine()

Move motor up in 0.5° steps

move_down_fine()

Move motor down in 0.5° steps

start_point_cloud_scan(*theta1*, *theta2*, *phi1*, *phi2*)

Move to coordinate arg1: elevation_min_scan arg2: elevation_max_scan arg3: azimut_min_scan arg4: azimut_max_scan

stop_point_cloud_scan()

Stop scan

get_point_cloud_scan_percent()

Point cloud scan status in perecent

class rubu.temas.**Common**(*port=8083*, *ip_address=''*)

Bases: object

This class included the common settings of Temas

shutdown()

Shutdown Temas

Returns
Shutdown temas

Return type
Boolean

restart()

Restart Temas

Returns
Restart temas

Return type
Boolean

get_temperature()

Read Temperature of Temas

Returns
Temperature Temas

Return type
String

get_ip()

Get ip address of Temas

Returns
Ip address

Return type
String

`get_sn()`

Get serial number of Temas

Returns
serial number is returned

Return type
String

`get_fw_version()`

Get firmware version of Temas

Returns
firmware version is returned

Return type
String

`get_hostname()`

Get hostname of Temas

Returns
Hostname is returned

Return type
String

`set_hostname(`*hostname*`)`

Set hostname of Temas

Returns
Hostname is changed

Return type
Boolean

`get_port_camera()`

Get port of Temas camera

Returns
Port from camera

Return type
String

`get_port_tof_camera()`

Get port of Temas tof camera

Returns
Port from tof camera

Return type
String

`get_port_common()`

Get port of Temas common

Returns
Port from Temas common

Return type
String

`get_port_control()`

Get port of control

Returns
Port from control

Return type
String

`get_ip_hosts()`

Get the host ip address of Temas

Returns
Ip adress of host

Return type
String

`get_mac()`

Get the mac address of Temas

Returns
Returns the mac address

Return type
String

`get_laser_x_los()`

Get the x value los of Laser in pixel

Returns
Returns the x value los

Return type
String

`get_laser_y_los()`

Get the y value los of Laser in pixel

Returns
Returns the y value los

Return type
String

`set_static_ip(ip, gateway, dns)`

static ip /24 dhcp arg1: ip-address arg2: gateway arg3: dns

`set_direct_static_ip(ip)`

static ip /24 without dhcp arg1: ip-address

`near_mode_on()`

tof near mode on

`near_mode_off()`

tof near mode off

set_temas_serialnumber(*sn*)

Set serial number of Temas

Returns
serial number is returned

Return type
String

CHAPTER

THREE

INDICES AND TABLES

- genindex
- modindex

PYTHON MODULE INDEX

r
rubu.temas, 3